Hands on Subnetting Practical

1. Subnetting Plan: 192.168.10.0/24 Divided into 4 Equal Subnets

- **Base Network:** 192.168.10.0/24
- **Subnets Required:** 444
- **Borrowed Subnet Bits (nnn):** 2n≥4 ⟹ n=2 bits2^n \ge 4 \implies n = 2\text{ bits}2n≥4⟹n=2 bits
- **New Prefix Length:** /24+2=/26/24 + 2 = \mathbf{/26}/24+2=/26
- **Subnet Mask:** 255.255.255.192\mathbf{255.255.255.192}255.255.255.192 (Binary: 11111111.11111111.11111111.11000000)
- **Block Size (Increment):** 256−192=64256 - 192 = \mathbf{64}256−192=64 addresses per subnet

Subnet Allocation Table

Subnet Name	Network Address	Subnet Mask	Usable Host IP Range	Broadcast Address
Subnet 1	192.168.10.0	255.255.255.192 (/26)	192.168.10.1 – 192.168.10.62	192.168.10.63
Subnet 2	192.168.10.64	255.255.255.192 (/26)	192.168.10.65 – 192.168.10.126	192.168.10.127
Subnet 3	192.168.10.128	255.255.255.192 (/26)	192.168.10.129 – 192.168.10.190	192.168.10.191
Subnet 4	192.168.10.192	255.255.255.192 (/26)	192.168.10.193 – 192.168.10.254	192.168.10.255

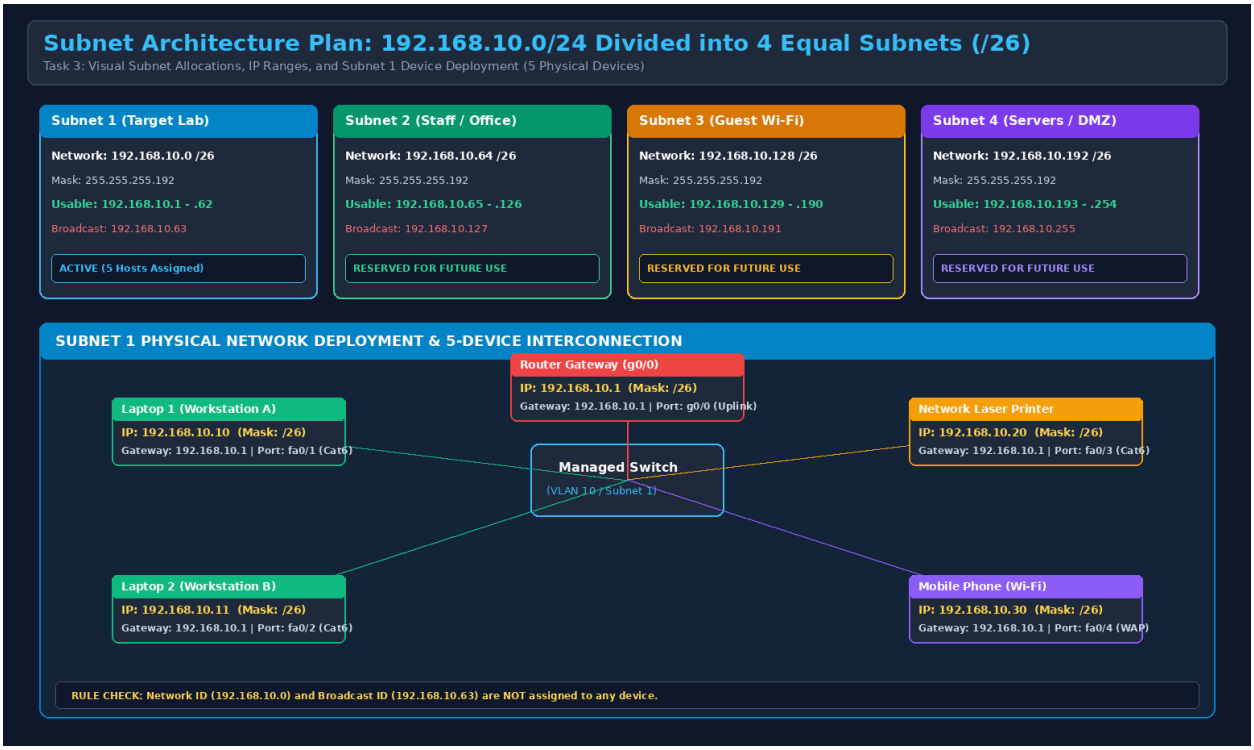
2. Device IP Assignment for Subnet 1 (192.168.10.0/26)

- **Usable Host Window:** 192.168.10.1 to 192.168.10.62
- **Reserved Addresses Not Used on Devices:**
    - Network Address: 192.168.10.0 (identifies the entire wire segment)
    - Broadcast Address: 192.168.10.63 (broadcasts to all hosts in Subnet 1)

Device	Assigned Static IP	Subnet Mask	Default Gateway	Connection Type / Port
Router (Default Gateway)	192.168.10.1	255.255.255.192	192.168.10.1	Router LAN Interface g0/0
Laptop 1 (Workstation A)	192.168.10.10	255.255.255.192	192.168.10.1	Switch Port fa0/1 (Cat6)
Laptop 2 (Workstation B)	192.168.10.11	255.255.255.192	192.168.10.1	Switch Port fa0/2 (Cat6)
Network Laser Printer	192.168.10.20	255.255.255.192	192.168.10.1	Switch Port fa0/3 (Cat6)
Mobile Phone	192.168.10.30	255.255.255.192	192.168.10.1	5 GHz Wi-Fi Access Point

3. Subnet Planning & Topology Diagram

The diagram below details the 4-subnet division across the network and maps the 5 connected devices in Subnet 1:



4. Step-by-Step Calculation Guide (Explaining to a Beginner)

Imagine your original network 192.168.10.0/24 is one large apartment building with **256 rooms** (numbered 0 to 255). You need to split this building into **4 equal, independent sections**:

Step 1: Borrowing Bits from the Host Portion

- In an IPv4 address, there are 32 bits total. A /24 prefix means the first 24 bits represent the **Network address**, leaving 32−24=8 bits32 - 24 = \mathbf{8\text{ bits}}32−24=8 bits for **Hosts**.
- With 8 bits, you have 28=2562^8 = 25628=256 total addresses.
- To create 4 subnets, we use the formula:

2n≥Number of required subnets2^n \ge \text{Number of required subnets}2n≥Number of required subnets

- Setting n=2n = 2n=2 gives 22=42^2 = 422=4. We borrow **2 bits** from the host portion and give them to the network portion.
- The new prefix becomes 24+2=/2624 + 2 = \mathbf{/26}24+2=/26.

Step 2: Calculating the New Subnet Mask

- In the 4th octet, we turn on the two highest bits:

Binary: 1 1 0 0 0 0 0 0\text{Binary: } \mathbf{1}\;\mathbf{1}\;0\;0\;0\;0\;0\;0Binary: 11000000

- Calculate the decimal value:

128+64=192128 + 64 = \mathbf{192}128+64=192

- The new subnet mask is:

255.255.255.192\mathbf{255.255.255.192}255.255.255.192

Step 3: Determining the Block Size (Increment)

- Subtract the last octet from 256:

256−192=64256 - 192 = \mathbf{64}256−192=64

- This means each subnet has **64 total IP addresses** and the subnets start at multiples of 64:
    - Subnet 1 starts at **.0**
    - Subnet 2 starts at **.64**
    - Subnet 3 starts at **.128**
    - Subnet 4 starts at **.192**

Step 4: Finding Usable Hosts per Subnet

- Remaining host bits: h=32−26=6 bitsh = 32 - 26 = \mathbf{6\text{ bits}}h=32−26=6 bits.

- We calculate usable hosts using the formula:

Usable Hosts=2h−2=26−2=64−2=62 usable hosts\text{Usable Hosts} = 2^h - 2 = 2^6 - 2 = 64 - 2 = \mathbf{62\text{ usable hosts}}Usable Hosts=2h−2=26−2=64−2=62 usable hosts

- **Why subtract 2?**
    1. The **first address** in every subnet is the **Network ID** (e.g., 192.168.10.0), which identifies the whole subnet.
    2. The **last address** in every subnet is the **Broadcast ID** (e.g., 192.168.10.63), used to send packets to all devices at once.
    3. Assigning either reserved address to a host device will cause operating system network configuration errors.

Step 5: Assigning Device Addresses

- In **Subnet 1**, usable IPs run from **192.168.10.1** to **192.168.10.62**.
- By convention, the very first usable IP (192.168.10.1) is given to the **Router** so other devices can reach the outside world.
- The remaining usable IPs (such as .10, .11, .20, and .30) are assigned to workstations, printers, and mobile phones, with each device sharing the exact same subnet mask (255.255.255.192) and default gateway (192.168.10.1).